Fiscal Year:	FY 2024	**Task Last Updated:**	FY 05/08/2024
PI Name:	Narayanan, Anand Ph.D.		
Project Title:	Partial Gravity and Sex-Difference Effects on the Venous Circulation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Immunology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	anarayan09@gmail.com	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	203 785 6055
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City:	Tallahassee	**State:**	FL
Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Former email address: anarayanan@fsu.edu		
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	06/01/2023	**End Date:**	05/31/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	13	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/31/2025 per NSSC information (Ed., 5/22/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Delp, Michael Ph.D. (Florida State University)		
Grant/Contract No.:	80NSSC23K0885		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA Space Biology aims to understand organism adaptations to the spaceflight environment. An understudied area of space biology is the spaceflight adaptation of the venous circulation. The venous circulation supports the cardiovascular system, and in turn all organs of the body, by transporting blood and its contents away from organ systems toward the heart and lungs. Consequently, venous return impacts cardiac function. As NASA aims to send astronauts to the Moon and Mars, understanding the adaptations of the venous circulation to microgravity, lunar gravity, and Martian gravity will not only increase our knowledge of cardiovascular function in space along a gravity continuum, but also uncover new potential astronaut medical risks. Furthermore, NASA will also be sending the first female astronaut into deep space as part of the Artemis Moon venture, and to date there exists no literature on possible sex differences of the venous circulation in response to spaceflight. This study will be the first comprehensive assessment of venous circulation adaptations to a simulated gravity continuum, and also investigate sex differences to increase our knowledge of male and female cardiovascular space biology adaptations. This study will investigate, for the first time, the functional, structural, and molecular adaptations of veins from multiple organ beds (head, heart, digestive system, and bone) to identify vascular biology, immunological, and local organ system physiological adaptations to simulated spaceflight environment conditions. In completing this study, we will have increased our overall knowledge and understanding of space biology, as well as identify prospective health risks of our astronauts in advance of their journeys into deep space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In completing these studies, I have shown how the vascular state of the venous circulation from different organ beds adapt to various spaceflight conditions (actual spaceflight, simulated microgravity and/or radiation, etc.). These studies will enhance our understanding of space biological adaptations of the cardiovascular system, and furthermore, increase our understanding of medical conditions astronauts develop while in space. In studying spaceflight biomedical adaptations, we also increase our knowledge of Earth-based biomedical science by investigating novel pathways activated by the unique spaceflight environment (i.e. gravity, low-dose radiation, isolation, etc.) that may provide new perspectives of how medical conditions develop and/or biology evolve on Earth. These studies provide foundational work that I will further characterize through various on-going and future projects.
Task Progress:	Project Status: The Early Career Investigator Award was tied with NASA Grant: 18-FG_team_2-0016: Effects of Simulated Microgravity and Partial Unloading on Organ Systems of the Body. [Ed. Note: See Grant #80NSSC19K1599; PI Michael Delp.] This project (18-FG_team_2-0016) has been delayed. Thus, the experimental work tied to this Award had been adapted for studies from Rodent Research-9 (Grant # NNX16AC28G) and Radiation, Simulated Weightlessness and Countermeasures: Effects on Cerebral and Coronary Vascular Function and Structure (Grant # 80NSSC19K1322). Aim 1 is ongoing, with 75% of experimental work completed; anticipating completion and submission of manuscripts, annual report, etc., by Fall 2024. Aim 2 is ongoing, with 75% of experimental work completed; anticipating completion and submission of manuscripts, annual report, etc., by Fall 2024.
Bibliography Type:	Description: (Last Updated: 05/22/2024)
Articles in Peer-reviewed Journals	Andrade MR, Azeez TA, Montgomery MM, Caldwell JT, Park H, Kwok AT, Borg AM, Narayanan SA, Willey JS, Delp MD, La Favor JD. "Neurovascular dysfunction associated with erectile dysfunction persists after long-term recovery from simulations of weightlessness and deep space irradiation." FASEB J. 2023 Dec;37(12):e23246. https://doi.org/10.1096/fj.202300506RR ; PMID: 37990646 , Dec-2023
Articles in Peer-reviewed Journals	Narayanan SA, Jamison Jr DA, Guarnieri JW, Zaksas V, Topper M, Koutnik AP, Park J, Clark KB, Enguita FJ, Leitão AL, Das S. "A comprehensive SARS-CoV-2 and COVID-19 review, Part 2: host extracellular to systemic effects of SARS-CoV-2 infection. " Eur J Hum Genet. 2024 Jan;32(1):10-20. https://doi.org/10.1038/s41431-023-01462-1 ; PMID: 37938797; PMCID: PMC10772081 , Jan-2024
Articles in Peer-reviewed Journals	Garcia-Medina JS, Sienkiewicz K, Narayanan SA, Overbey EG, Grigorev K, Ryon KA, Burke M, Proszynski J, Tierney B, Schmidt CM, Mencia-Trinchant N. "Genome and clonal hematopoiesis stability contrasts with immune, cfDNA, mitochondrial, and telomere length changes during short duration spaceflight." Precis Clin Med. 2024 Apr 8;7(1):pbae007. https://doi.org/10.1093/pcmedi/pbae007 ; PMID: 38634106; PMCID: PMC11022651 , Apr-2024